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Adoption of Generative AI (Gen-AI) In Sub-Saharan Africa: Extension of the UTAUT Model

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ABSTRACT

AI can improve intelligence and capabilities, paving the door for transformative outcomes in Africa's socioeconomic revolution. The Unified Theory of Acceptance and Use of Technology (UTAUT) can be a useful framework for investigating the adoption of generative artificial intelligence (Gen-AI) in Sub-Saharan Africa. UTAUT analyzes a variety of elements that influence technology acceptance and use, such as performance expectancy, effort expectancy, social influence, and facilitating conditions. These constructs can shed light on the drivers and challenges affecting the application of generative AI in Africa. To support Jain's model, authors also argue that algorithmic aversion mediates the influence of performance expectancy and the use of AI. An online survey of African-based technology professionals using AI-enabled tools has been distributed and we present initial results in this paper. We offer focused techniques to utilize generative AI's revolutionary potential for innovation, economic growth, and social development in Africa based on an in-depth examination of the constraints and opportunities.

Keywords

Gen-AI, UTAUT, Sub-Saharan Africa.

INTRODUCTION

Artificial intelligence (AI), particularly machine learning (ML), has made tremendous advances in data analysis and computing in recent years. These advancements have enabled programs to operate intelligently, demonstrating the ability to grasp and effectively process data (Sarker, 2021). Generative AI (Gen-AI) has emerged as a transformational technology with the potential to disrupt numerous elements of the global economy and African society (Ferrag et al., 2023). Several prominent corporations have made substantial contributions to the field of artificial intelligence (AI) research, which has revolutionized various industries around the world. OpenAI (OpenAI et al., 2021), Google (Lin, 2023), Microsoft (Brühl, 2023), Meta, Deepmind (Goldstein et al., 2023), Nvidia (Khalifa et al., 2023), IBM (Townsend & Hunt, 2019), and many others are prominent players in creating generative AI.

OpenAI, a major organization dedicated to ensuring that artificial intelligence benefits all of humanity (ShneidermanBen, 2020), has been at the forefront of cutting-edge research, pioneering advancements in natural language processing and generative models. Google, a digital behemoth, has made amazing advances in AI research with projects such as Google Brain (Sajjadi et al., 2018) and DeepMind (Serrano-Muoz et al., 2022), pushing the boundaries of machine learning and neural networks. Microsoft (Dao, 2023), another significant competitor, has made significant investments in AI research, employing cutting-edge approaches in fields such as computer vision (Chociej et al., 2019) and speech recognition (COA, 2020). These firms, along with many others, have played an important role in defining the AI landscape and have enormous promise for unlocking new opportunities for individuals, enterprises, and governments worldwide (Mhlanga, 2023; Salvaris et al., 2018).

In Sub-Saharan Africa, the use of AI-enabled tools is evolving at a very slow pace, millions of people are not even aware of Generative AI tools (Floyd, 2023). From the healthcare sector to education, agriculture, and infrastructure, Gen AI tools have rapidly become consistent in the lives of citizens. Although through the United Nations Sustainable Development Goals, AI is beginning to make some level of impact, the majority of Sub-Saharan African nations currently lack comprehensive regulatory frameworks that would effectively encourage the development and collection of high-quality driven data, govern AI-driven business models, and incentivize ethical AI.

Considering the above, this article aims to answer the following questions: (1) How can Generative AI be harnessed to address societal, economic, and cultural challenges in Africa, considering local contexts, ethical implications, and the potential for fostering innovation and sustainable development; (2) What stimulates the interests of Sub-Saharan African residents and

Nationals to use Gen-AI tools? (3) Can we use UTAUT to explain the adoption of Gen-AI tools in sub-Saharan Africa? (4) What role does Algorithmic Aversion play in the adoption of Gen-AI?

The reluctance towards adopting AI can stem from its distinctive nature compared to other types of technology (Jain et al., 2022). (Venkatesh, 2022) emphasizes the importance of identifying unique factors that influence the adoption of AI-based technology, given its emerging form. Scholars have been pushed to think distinctly about traditional technology adoption models to consider the contextual factors and characteristics of these new technologies (Hong et al., 2014; Dwivedi et al., 2019; Dwivedi et al., 2020). AI aversion can stem from its uniqueness. Thus, there is a need to look beyond existing models to see its effects on users in Sub-Saharan Africa. Extending the UTAUT model will directly or indirectly increase our understanding of this technological and social phenomenon (Makarius et al., 2020)

UNIFIED THEORY OF ACCEPTANCE AND USE OF TECHNOLOGY

The Unified Theory of Acceptance and Use of Technology (UTAUT) (Williams et al., 2015) is a well-known theoretical framework that seeks to comprehend and explain people's acceptance and use of technology. UTAUT was developed by Venkatesh et al. (2003) to provide a thorough understanding of technology adoption behavior. It incorporates and expands numerous previous theories, including the Technology Acceptance Model (TAM) and the Theory of Reasoned Action (TRA).

(Cao et al., 2021; Jain et al., 2022) extended the approach by integrating an integrated AI acceptance-avoidance model in their study as seen in Figure 1. This expanded model considers both positive and negative aspects that influence managers' attitudes and behaviors toward the deployment of AI. Previous research has identified technology's detrimental consequences (Vinuesa et al., 2020; Floridi & Cowls, 2022 & Vimalkumar et al., 2021). As a result, when creating a model, it is critical to account the unfavorable impressions of AI (Davenport et al., 2020; Wang & Siau, 2019 & Westerlund, 2019).

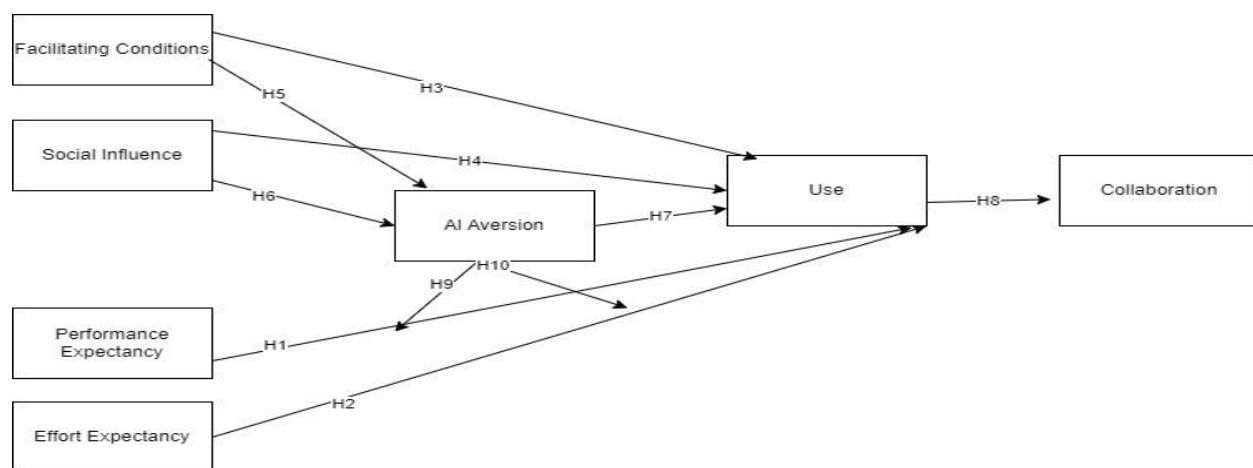


Figure 1. Proposed Research Model (Jain et al., 2022)

UTAUT identifies four key factors that influence individuals' acceptance and use of technology:

Performance Expectancy

This factor examines how strongly people believe that employing a specific technology would improve their performance and help them reach their goals. It emphasizes on the perceived benefits and expected positive results of technology use (Whittaker et al., 2018; Ain et al., 2016 & Malik et al., 2019). Although AI has been shown to improve performance expectancies, there is little empirical evidence of the role AI aversion plays in its adoption (Jain et al., 2022). Owing to these, we propose the following hypotheses:

H1: Performance expectancy will positively affect the use of Gen-AI in Sub-Saharan Africa

H9: AI aversion will mediate the influence of performance expectancy and the use of Gen-AI in Sub-Saharan Africa

Effort Expectancy

The perceived ease of use and utility of technology is reflected in effort expectancy. It takes into account elements such as the complexity of learning and using technology, as well as individuals' confidence in their capacity to do so (Rahi et al., 2019;

Ragheb et al., 2022; Ain et al., 2016). Trust difficulties and misunderstandings can stymie AI adoption in Sub-Saharan Africa. In light of this, the hypotheses listed below were formed.

H2: Effort Expectancy positively influences the use of Gen-AI tools in Sub-Saharan Africa

H10: AI aversion will moderate the impact of effort expectation and the adoption of Gen-AI-enabled gadgets in Sub-Saharan Africa.

Social Influence

Social influence research looks at the impact of social norms, subjective norms, and other people's opinions and recommendations on individuals' technology adoption decisions (Manca et al., 2019; Beldad & Hegner, 2018; Hong & Cho, 2023). It investigates the impact of social elements, such as peers, superiors, and influential individuals, on changing people's views and intentions toward technology, particularly AI. This led to the hypotheses below:

H4: The use of Gen-AI tools in Sub-Saharan Africa is influenced positively by social influence.

H6: Social influence influences algorithmic aversion in Sub-Saharan Africa.

Facilitating Conditions

The availability of appropriate resources, support, and infrastructure to assist technology adoption is referred to as facilitating conditions (Kumar & Ayedee, 2021 & Grover et al., 2022). It takes into account things like technical support, training, corporate policies, and interoperability with existing systems. In Sub-Saharan Africa, government assistance for information technology has a multiplier effect on the use of information technology tools (AbdulKareem et al., 2020; Beda, 2019). This assistance has taken the shape of facilitating conditions. Today, policies in the form of supportive conditions for AI tools are being implemented (Sy et al., 2019). This led to the following hypotheses:

H3: Facilitating conditions have a positive impact on the usage behavior of Gen-AI products in Sub-Saharan Africa.

H5: Facilitating conditions have an effect on algorithmic aversion in Sub-Saharan Africa.

Algorithmic Aversion

This refers to people's reluctance or skepticism in relying on or accepting algorithmic decision-making systems (Hobson et al., 2023 & Oleksy et al., 2023). It derives from a lack of trust, concerns about openness, and apprehension about potential biases or errors in algorithmic processes (Mahmud et al., 2023). Algorithmic aversion can appear in a variety of scenarios, including the employment of AI tools in hiring processes, loan approvals, and recommendation systems (Bogert et al., 2021 & Berger et al., 2021).

Due to perceived limitations or ethical problems linked with algorithmic decision-making, individuals may choose human judgment or traditional decision-making methods over algorithms (Cao et al., 2021). The necessity of addressing transparency, fairness, and accountability (Hobson et al., 2023) in the design and deployment of algorithmic systems (Raji, 2020) to encourage confidence and acceptance (Cabiddu et al., 2022) among users is highlighted by algorithmic aversion. Based on this, we hypothesize that

H7: The use of Gen-AI tools is influenced by Algorithmic Aversion in Sub-Saharan Africa.

UTAUT (Gunasinghe & Nanayakkara, 2021) acknowledges that the influence and significance of these aspects varies among contexts, user groups, and technologies. As a result, the framework allows for the incorporation of new moderators (Celik, 2016) to help explain the adoption behavior.

UTAUT gives useful insights into the elements influencing individuals' adoption (Akinuwaesi et al., 2016) and usage of technology by taking these four major factors into account. It has been frequently used to assist system design (Williams et al., 2015), and inform methods for fostering successful technology implementation (Maghazei et al., 2022) and utilization (Beaman et al., 2021).

Collaboration

Collaborative tools have been used to test the collaborative climate amongst teams and management of tasks within industries and organizations (Larson & DeChurch, 2020, Isaac et al., 2019, 2019), however, there is little to no empirical literature that assesses the collaborative experience of residents with the use of AI-enabled tools.

H8: The use of Gen-AI tools in Sub-Saharan Africa positively influences collaboration amongst residents.

RESEARCH DESIGN

This study proposes the use of the UTAUT model to test the adoption of Gen-AI tools, of course with the addition of algorithmic aversion which adds value to the model. We believe that this model will help us understand the current state of Gen-AI tools adoption in Sub-Saharan Africa. We are currently in the process of data collection from technology professionals in Nigeria, Cameroon, and Uganda whose responses will allow us to strengthen our proposed model. To the best of our knowledge, this is the first time this proposed model is being used to test the adoption of Gen-AI in sub-Saharan Africa. People play a significant role in the acceptance of new digital tools. In future work, we plan to conduct a user survey on what Gen-AI tools are used by residents across the different sectors from education, government, business, and healthcare. The survey could also identify current challenges faced by residents.

As shown in Figure 2 below, preliminary data collection includes 54 technology professionals from Nigeria, Cameroon, and Uganda. Of those 57, majority believe AI is easy to use and makes life better, AI also enhances communication and decision making.

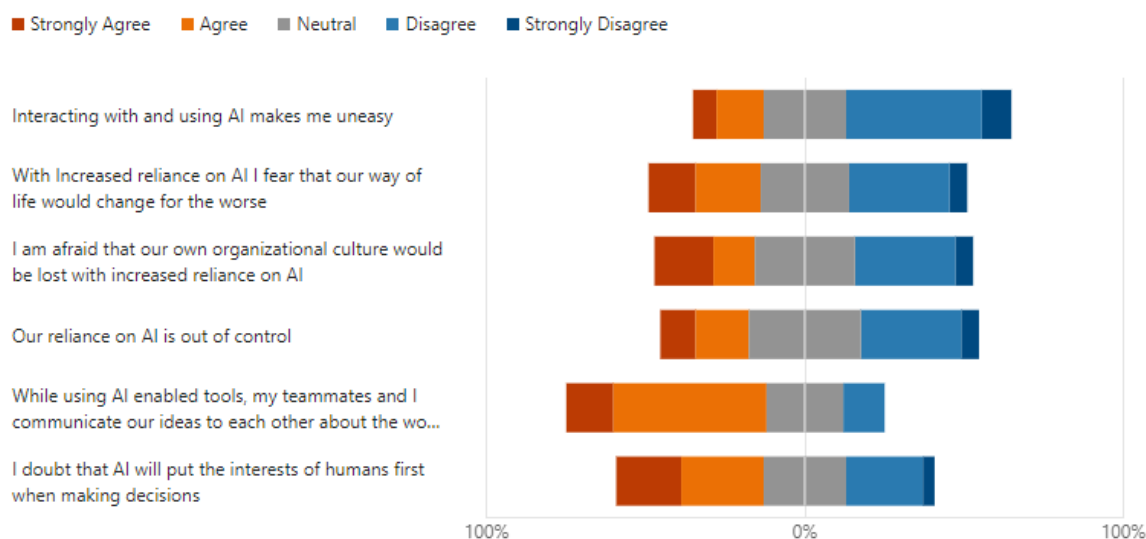


Figure 2. Survey Question on AI Aversion

CONCLUSION

Finally, this study underscores the revolutionary potential of generative AI in Africa, as well as the necessity for concentrated efforts to fully realize its benefits. Africa can use generative AI to drive innovation, economic growth, and societal improvement by tackling issues and executing smart actions. Africa stands to profit from advances in generative AI, and by encouraging collaboration, investment, and skill building, we can harness its power for the continent's broader socioeconomic growth.

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